long term (invariant measure) conservation properties; in this way they can provide a rational and flexible framework for designing stable algorithms.

Once a stochastic molecular dynamics-based framework is developed it is possible to modify it or use it as a building block in order to enhance sampling efficiency. One can for example incorporate these methods into *kinetic Monte-Carlo*, other metropolized schemes, or various advanced "path sampling" techniques. Thus the design of good molecular SDE methods is the cornerstone of much of modern molecular modelling.

7.1 Bias

In this chapter, the focus is on sampling strategies based on approximating Langevin dynamics (6.32)–(6.33). As will be seen, these methods offer a great deal of flexibility and, most importantly, reliability and robustness in the approximation of the canonical measure. All standard methods based on discretizing SDEs introduce some stepsize-dependent error in computed averages, which we shall refer to as bias (related to the truncation error of the discretization scheme). This bias can be corrected for using certain 'rebiasing' algorithms, for example using a Metropolis (accept-reject) condition [44, 47], however there may be significant computational overhead in employing these schemes leading to inefficiency compared to standard MD, in spite of the introduced error.

An illustration of the sampling bias (i.e., due to discretization error) is shown in Fig. 7.1. As the stepsize is increased, the error in sampling is increased as well, limiting the effectiveness of numerical methods. This bias can be dramatically different for different numerical methods. As we shall show, with the right choice of numerical method it is often possible to substantially reduce this error, and it is also possible to calculate (under some assumptions) the perturbation introduced by the numerical method, and to correct for its presence.

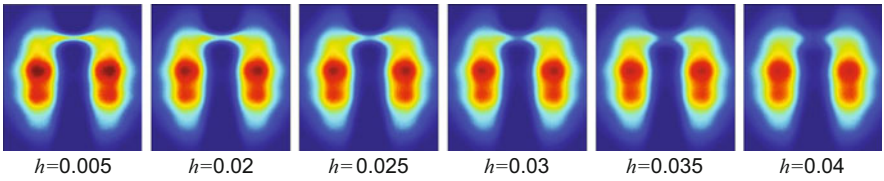


Fig. 7.1 We illustrate computed (canonical) distributions of $\boldsymbol{q} = (x, y)$ for a planar potential energy function with two distinct basins connected by a narrow transition corridor, where high/low intensity is colored *red/blue* respectively. Each distribution is computed through simulations using a different stepsize parameter—no other properties are changed between the simulations. The differences between the distributions are the result of discretization error. At simulations with a larger timestep we can see a qualitative change in the sampled landscape, with the bridge between the two regions disappearing. Figure reprinted with permission from [222]. Copyright 2013, AIP Publishing LLC